

MTH204

Linear Algebra

Course Textbook

Department of Mathematical Sciences
UNIST

Linear Algebra

Course Textbook

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \vee Q) \wedge R$$

is evaluated by first computing $P \vee Q$.

- The **logical connectives** are interpreted in the order

$$\neg > \wedge > \vee > \implies > \iff$$

Hence

$$\neg P \wedge Q \vee R \implies S \iff T$$

is understood as

$$(((\neg P) \wedge Q) \vee R) \implies S) \iff T$$

- A **quantifier** binds the entire formula that follows it. Thus

$$\forall x, P(x) \wedge Q(x) \implies R(x)$$

means

$$\forall x ((P(x) \wedge Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

$$\begin{array}{c} P \\ \Downarrow \\ Q \end{array}$$

merely expresses that P is true, and therefore Q is true in the current discussion; it is not itself a proposition.

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Vector Spaces

1.1 Definitions

1.2 Bases

1.3 Dimension of a Vector Space

1.4 Sums and Direct Sums

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Matrices

2.1 The Space of Matrices

2.2 Linear Equations

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Chapter 3

Linear Mappings

3.1 Mappings

3.2 Linear Mappings

3.3 The Kernel and Image of a Linear Map

3.4 Composition and Inverse of Linear Mappings

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Determinants

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Eigenvectors and Eigenvalues

8.1 Eigenvectors and Eigenvalues

8.2 The Characteristic Polynomial

8.3 Eigenvalues and Eigenvectors of Symmetric Matrices

8.4 Diagonalizations of a Symmetric Linear Map

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Polynomials and Matrices

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Chapter 10

Triangulation of Matrices and Linear Maps

10.1 Existence of Triangulation

10.2 Theorem of Hamilton-Cayley

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Polynomials and Primary Decomposition

11.1 The Euclidean Algorithm

11.2 Greatest Common Divisor

11.3 Unique Factorization

11.4 Application to the Decomposition of a Vector Space

11.5 Schur's Lemma

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Chapter 12

Convex Sets

12.1 Definitions

12.2 Separating Hyperplanes

12.3 Extreme Points and Supporting Hyperplanes

12.4 The Krein-Milman Theorem

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